

Threshold Forecasting Under Rare Events

Early KPI Forecasting with Survival Curves,
Bayesian Updating, and Monte Carlo
Simulation Applied to Mini Golf Milestones



Forecasting Rare Milestones Before They Happen

Rare events can be forecast early, but only if uncertainty is modeled, not ignored.

- Survival analysis and Bayesian updating reveal milestone timing long before outcomes are fixed.
- These same methods apply directly to early KPI forecasting in business contexts.



Past performance anchors the forecast.

- Five completed seasons define when and how often the “break 30” milestone has occurred.
- Pooled priors provide a baseline for new course forecasts, capturing both variability and rarity.

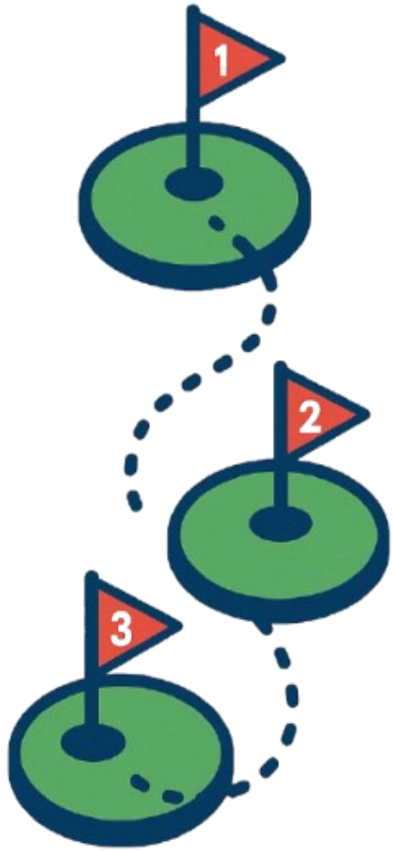


Early insights drive faster, smarter decisions.

- Forecasts evolve as new data arrives, guiding adjustments before results are final.
- Turning uncertainty into probability enables proactive resource allocation.



Roadmap



Historical Benchmarks

- Prior seasons provide reference points for how often rare milestones have been reached.
- These outcomes anchor a statistical baseline before any live data is introduced.



Live Forecast with Early Data

- Partial in-season results are incorporated as they arrive, refining expectations in real time.
- Survival analysis, Bayesian updating, and Monte Carlo simulation combine to generate credible early signals.



Business Implications

- Rare-event forecasting delivers actionable insight before outcomes are known.
- The parallels between “breaking 30” and hitting KPI thresholds show how methods transfer beyond sports.

The “Breaking 30” Mini Golf Challenge

- The Sanicki brothers attempt to finish 18 holes in 29 strokes or fewer, with only one round per day, create a naturally sparse stream of data.
- Each new course functions as a fresh “season,” meaning past outcomes provide only weak guidance for future ones.
- Results arrive sequentially, reflecting the same forecasting challenge as early business KPI thresholds.



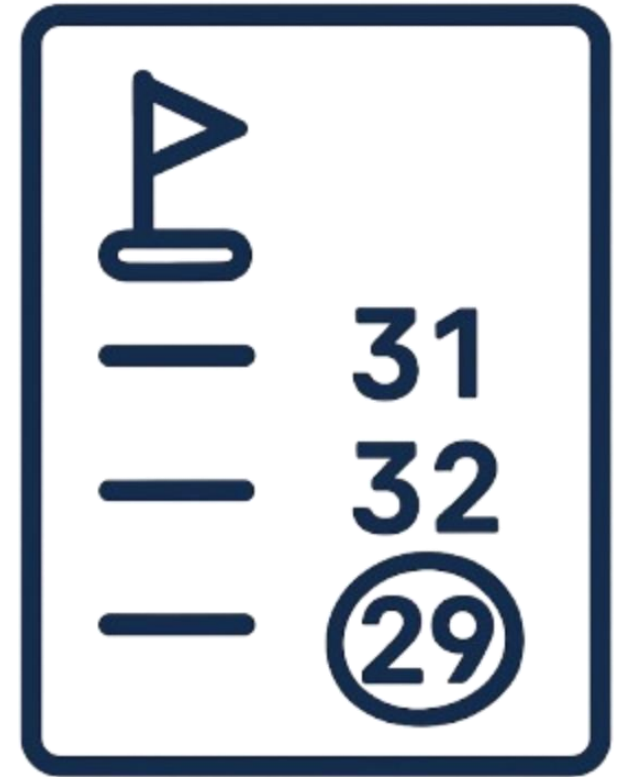
The Forecasting Challenge: Certainty vs. Timing

Early	Naïve Guesses • Predictions made with minimal data • Heavily influenced by noise or chance • Difficult to trust for real decisions • Creates false confidence early in the process	The Ideal Zone • Forecasts made before the outcome is fixed • Uses historical priors + live updates • High enough certainty to guide action • Rarely achievable without advanced methods
	The Dead Zone • Predictions delayed yet still uncertain • Adds no strategic value to decision-making • Wastes time and resources • Easily dismissed by stakeholders	Perfect Hindsight • Outcome known with full certainty • Accurate, but only after the fact • Provides no chance to adjust strategy • Reporting, not forecasting
Low Confidence		High Confidence

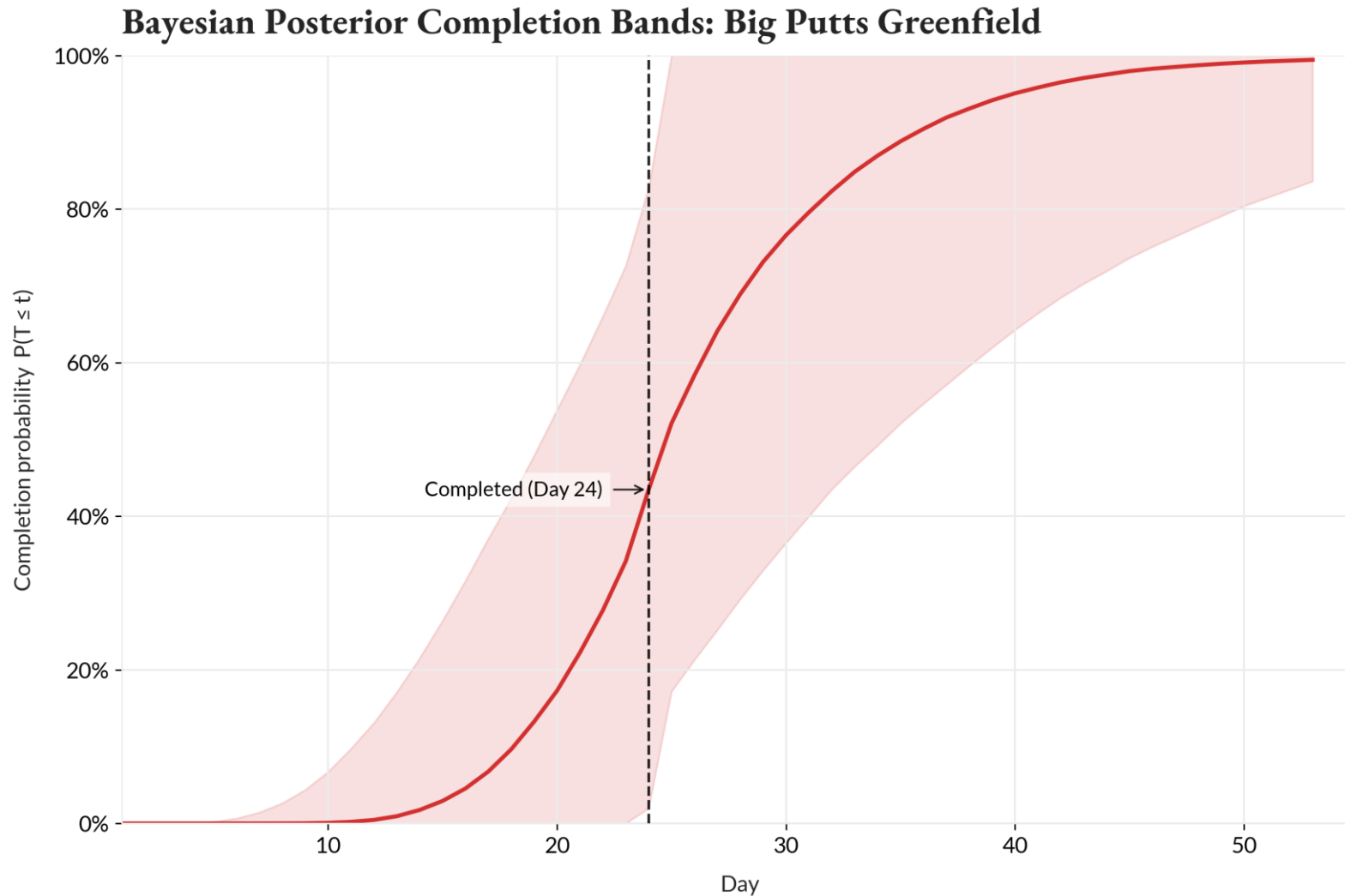
01

Historical Benchmarks

What past seasons tell us about milestone timing and variability.



Season 1: Big Putts Greenfield



Insights

How to read this chart

- The red curve is a Bayesian estimate of completion probability over time.
- The shaded band shows the 90% credible forecast range.

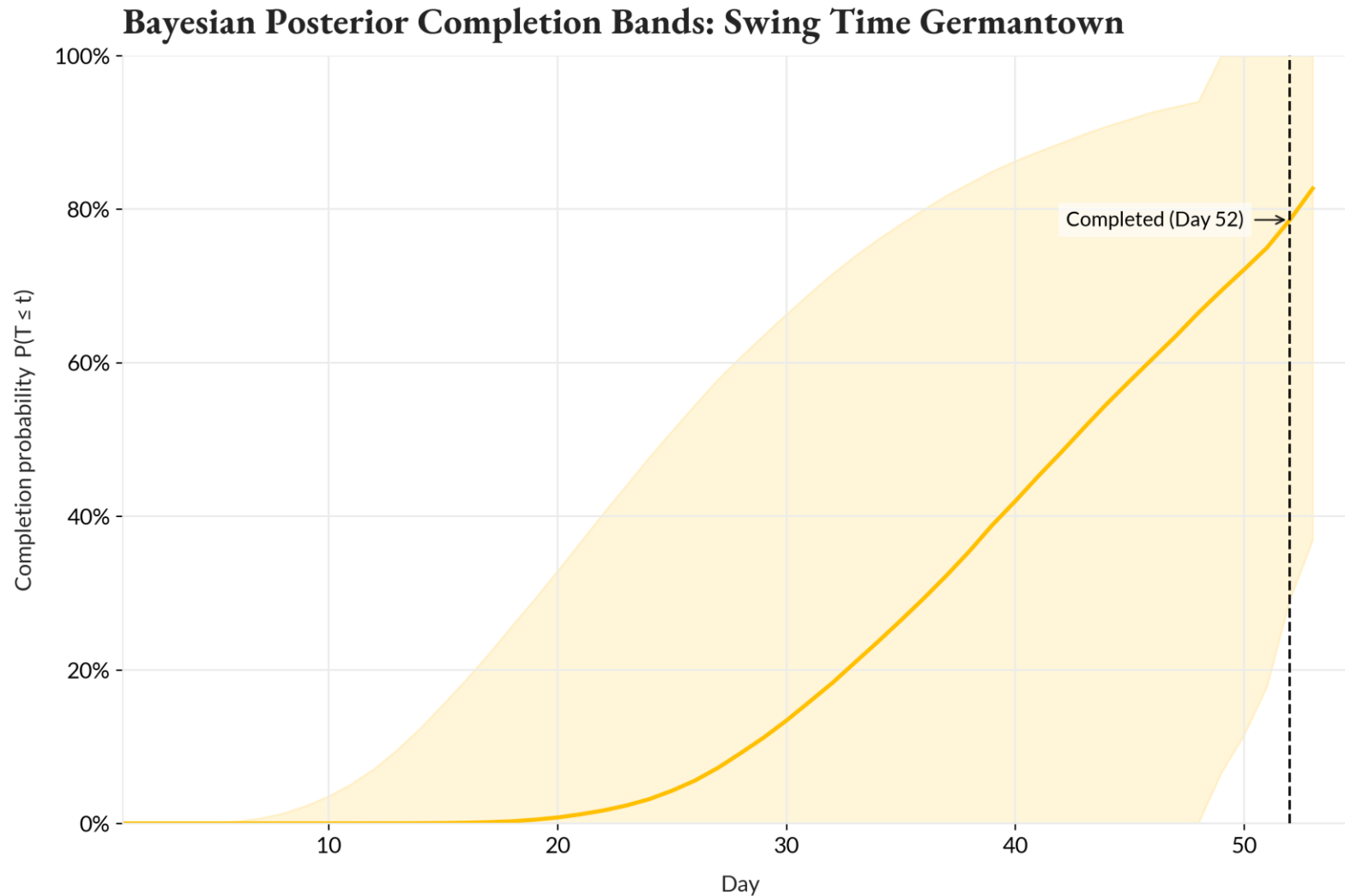
Season 1 pattern

- Completion came on Day 24, almost exactly matching the forecast median (24.7).
- At that point, success probability was ~43%.

Role in analysis

- Provides the first benchmark profile of milestone timing at an individual course.
- Serves as the most accurate prior, grounding the pooled analysis.

Season 2: Swing Time Germantown



Insights

Observed outcome

- Completion finally came on Day 52, the longest run in the dataset.
- Progress always seemed possible but never assured until the end.

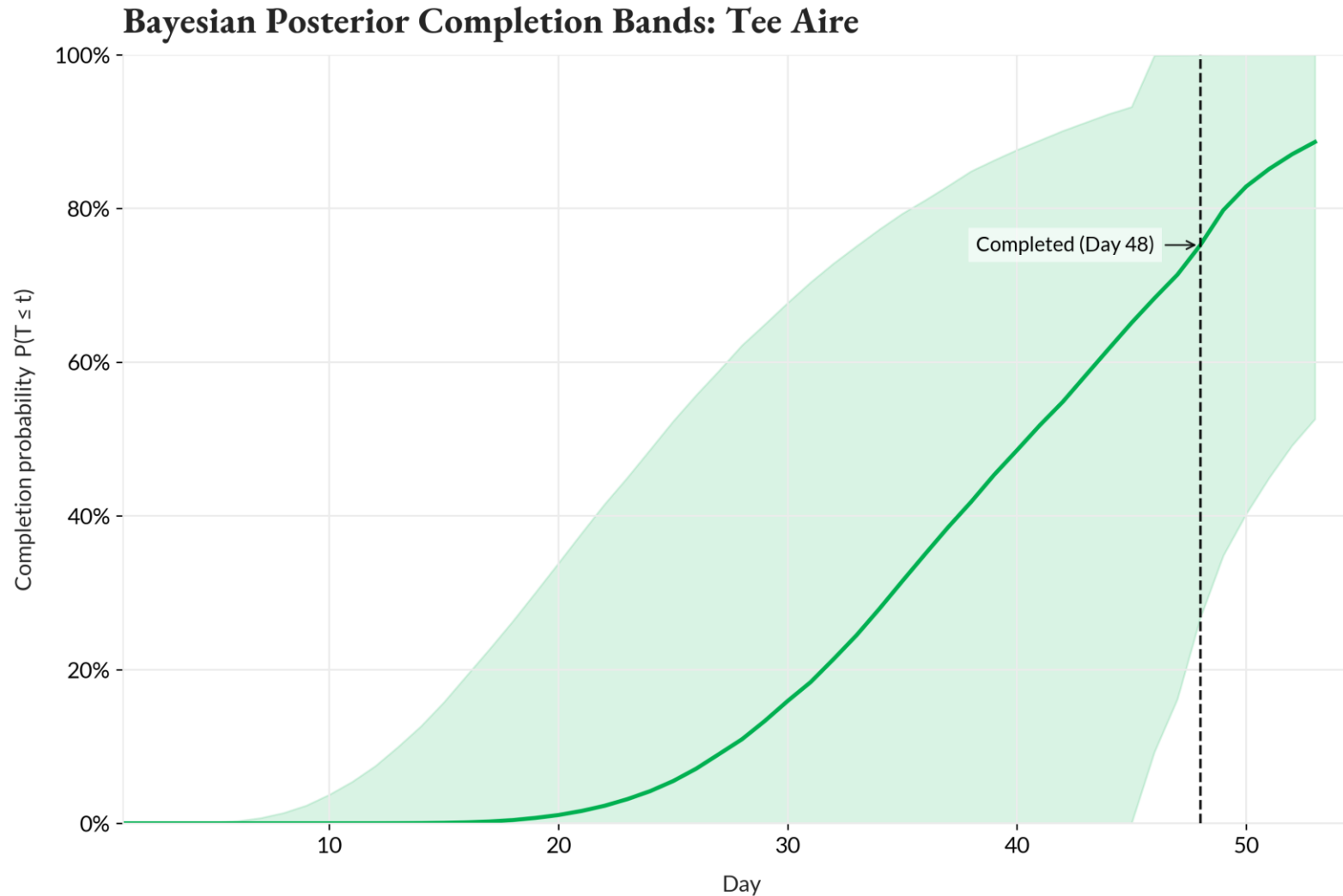
Forecast performance

- Median forecast was about 42.6 days, earlier than the observed outcome.
- By Day 40, the model estimated about 42 percent chance of success, leaving the result uncertain.

Analytical role

- Illustrates a season defined by suspense rather than acceleration.
- Balances pooled priors toward slow building completions.

Season 3: Tee Aire



Insights

Observed outcome

- Completion came on Day 48, one of the slowest seasons in the dataset.
- Probability climbed gradually, never spiking toward certainty.

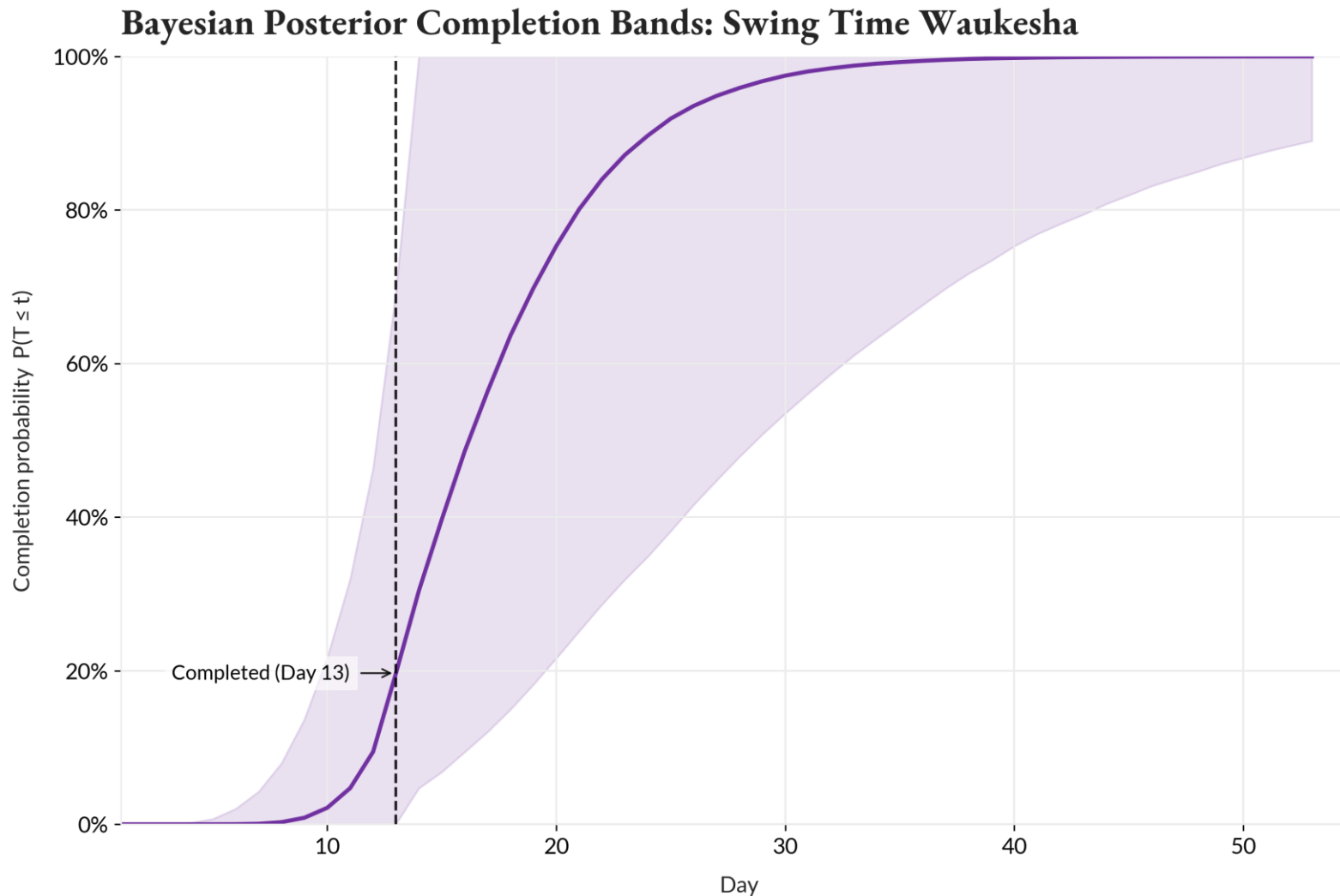
Forecast performance

- Median forecast was about 40.4 days, undershooting the observed result.
- Actual completion still fit inside the 90 percent credible interval of 20 to 54 days.

Analytical role

- Captures the grind of late completions without sharp breakthroughs.
- Pulls pooled priors toward slower outcomes.

Season 4: Swing Time Waukesha



Insights

Observed outcome

- Completion achieved on Day 13, the fastest season in the dataset.
- High signs of course wear created easier lines, helping probability rise to ~50% by Day 15.

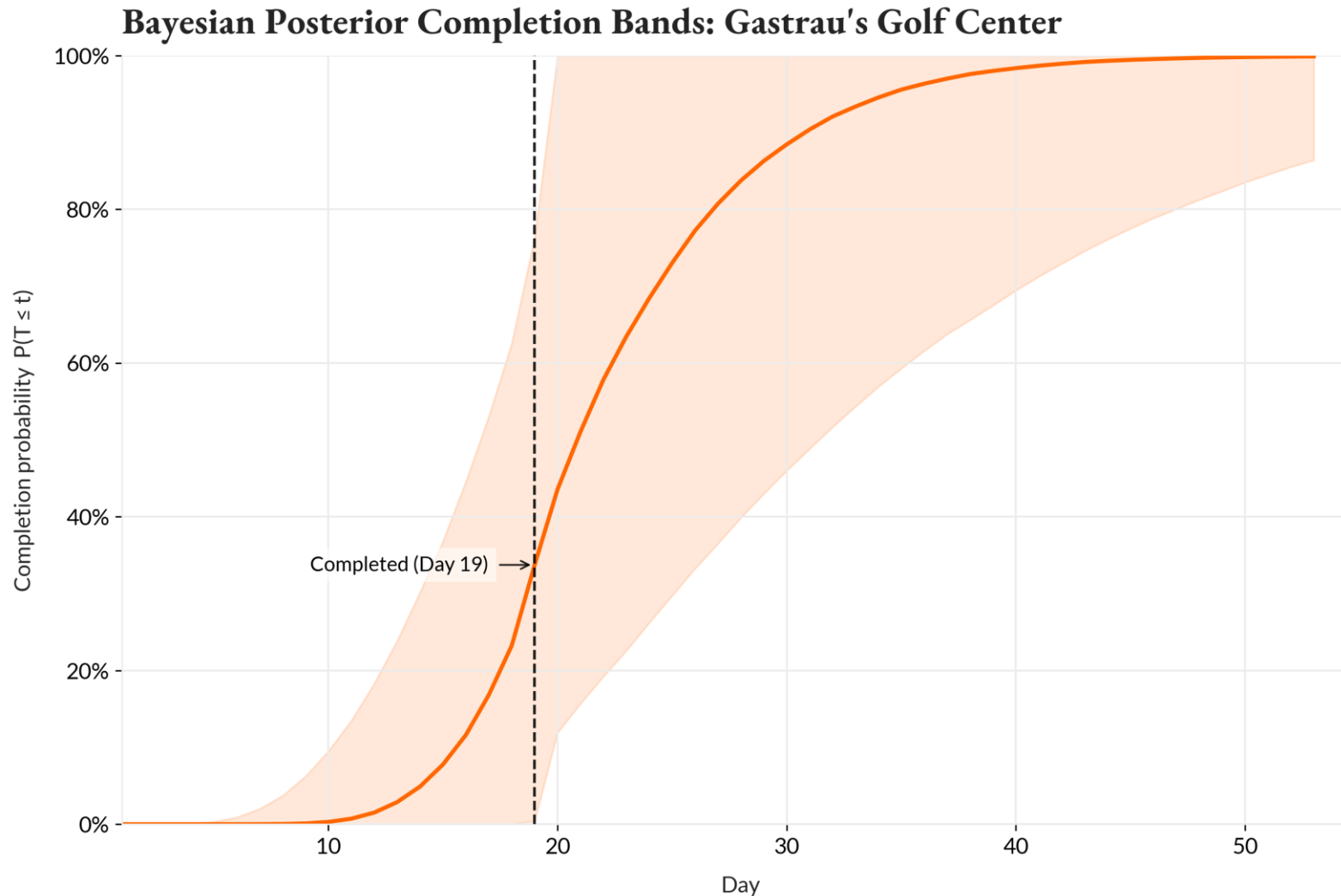
Model alignment

- Median forecast was about 16 days, slightly later than observed.
- Actual result landed comfortably inside the 90 percent interval of 10 to 30 days.

Analytical role

- Captures the early-bound case of milestone timing.
- Balances pooled priors by anchoring the fast edge of observed outcomes.

Season 5: Gastrau's Golf Center



Insights

Observed outcome

- Completion came on Day 19, near the forecast median of about 21 days.
- Early rounds were difficult, with low success probability in the first two weeks.

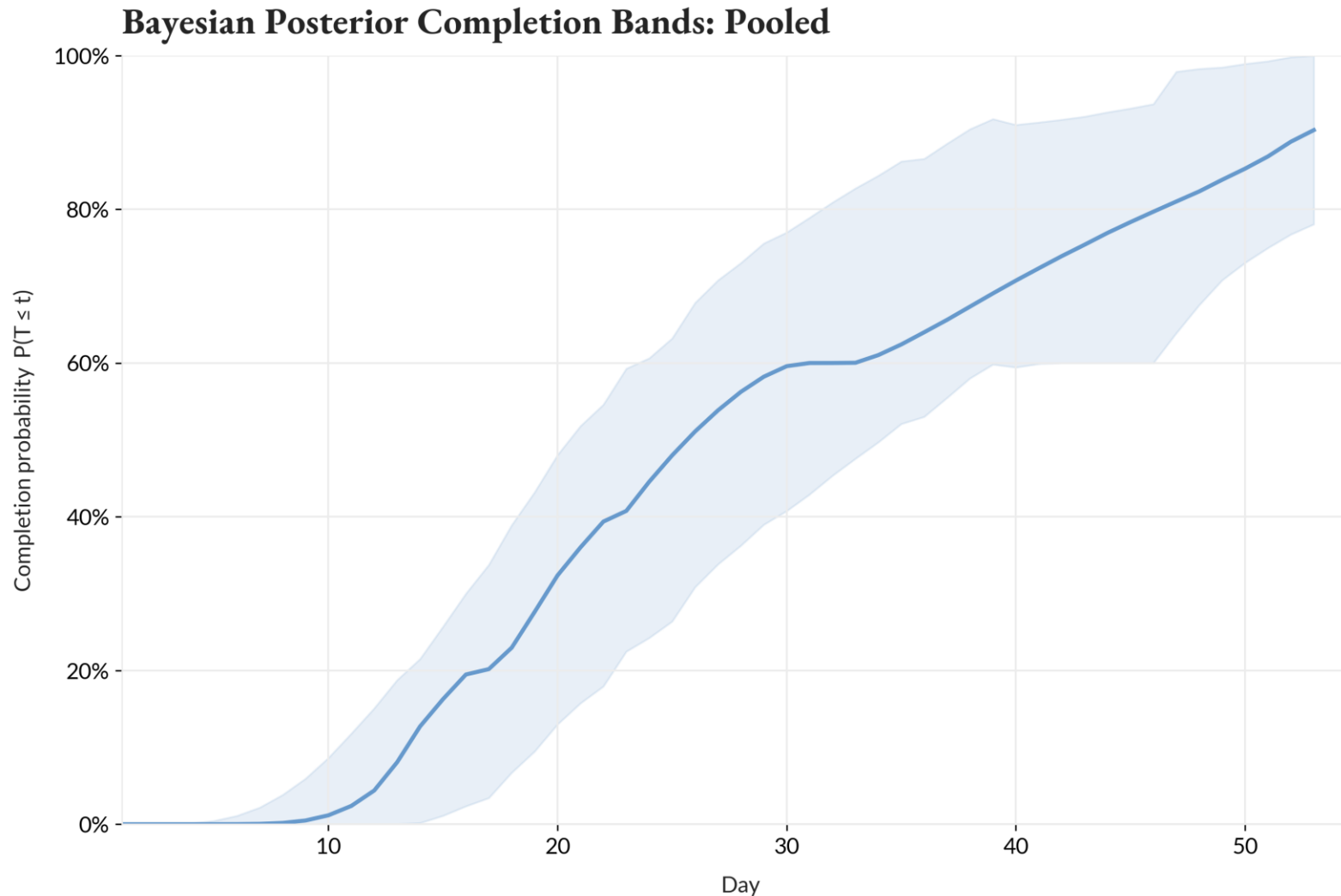
Performance trajectory

- By Day 15, success probability was only about 14 percent but then rose sharply.
- Improvement reflected skill adaptation rather than randomness or luck.

Analytical role

- Shows how consistent play on a predictable course can drive steady improvement.

Pooled View Across All Courses



Insights

Observed pattern

- Median pooled forecast placed completion around Day 26.
- The curve rose smoothly, blending across very different course profiles.

Forecast performance

- The 90 percent credible interval spanned roughly Day 21 to Day 34.
- Only Big Putts landed inside this range; others were earlier or later, reflecting course-specific difficulty.

Analytical role

- Highlights that deviations are expected when aggregating unique courses.
- Serves as the central prior, showing the “typical” trajectory despite variation.

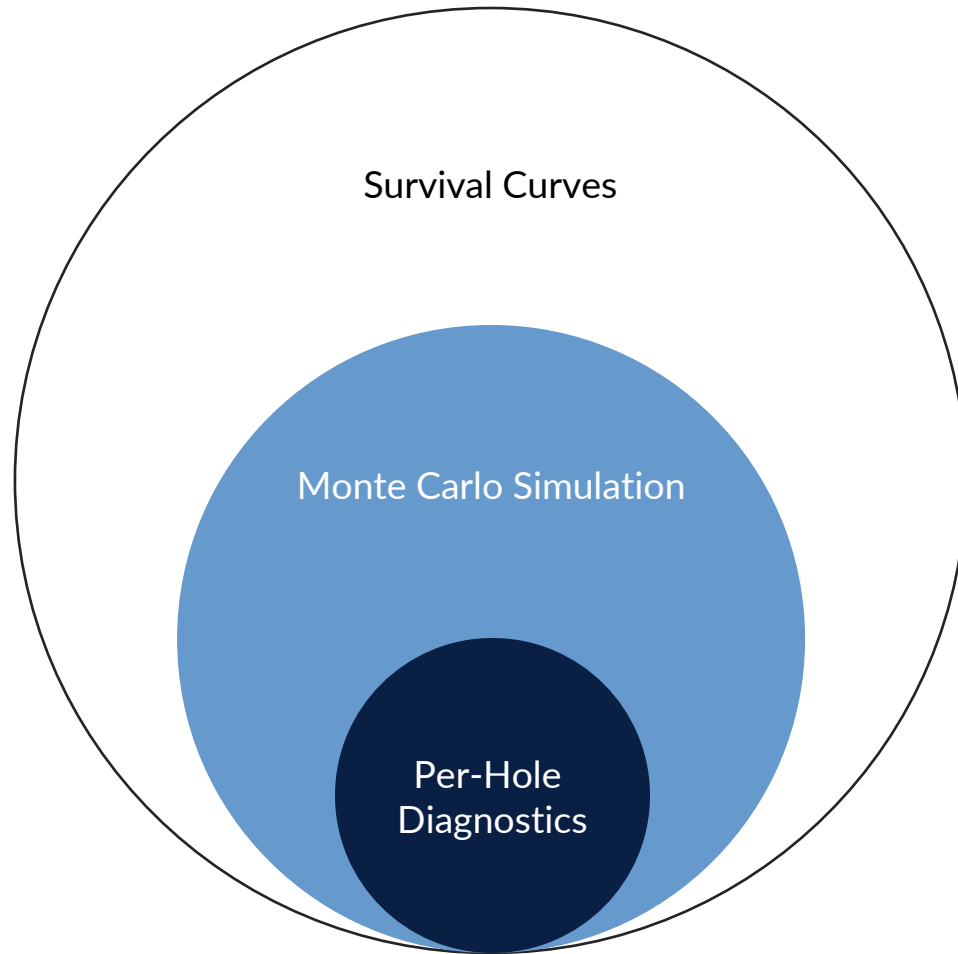
02

Live Forecast at Day 10

Turning partial results into actionable probability estimates.



The Forecasting Toolkit: Three Lenses on Uncertainty



Survival Curves

- Shows how the odds of breaking 30 build day by day.
- Gives the broad picture of progress and timing.

Monte Carlo Simulation

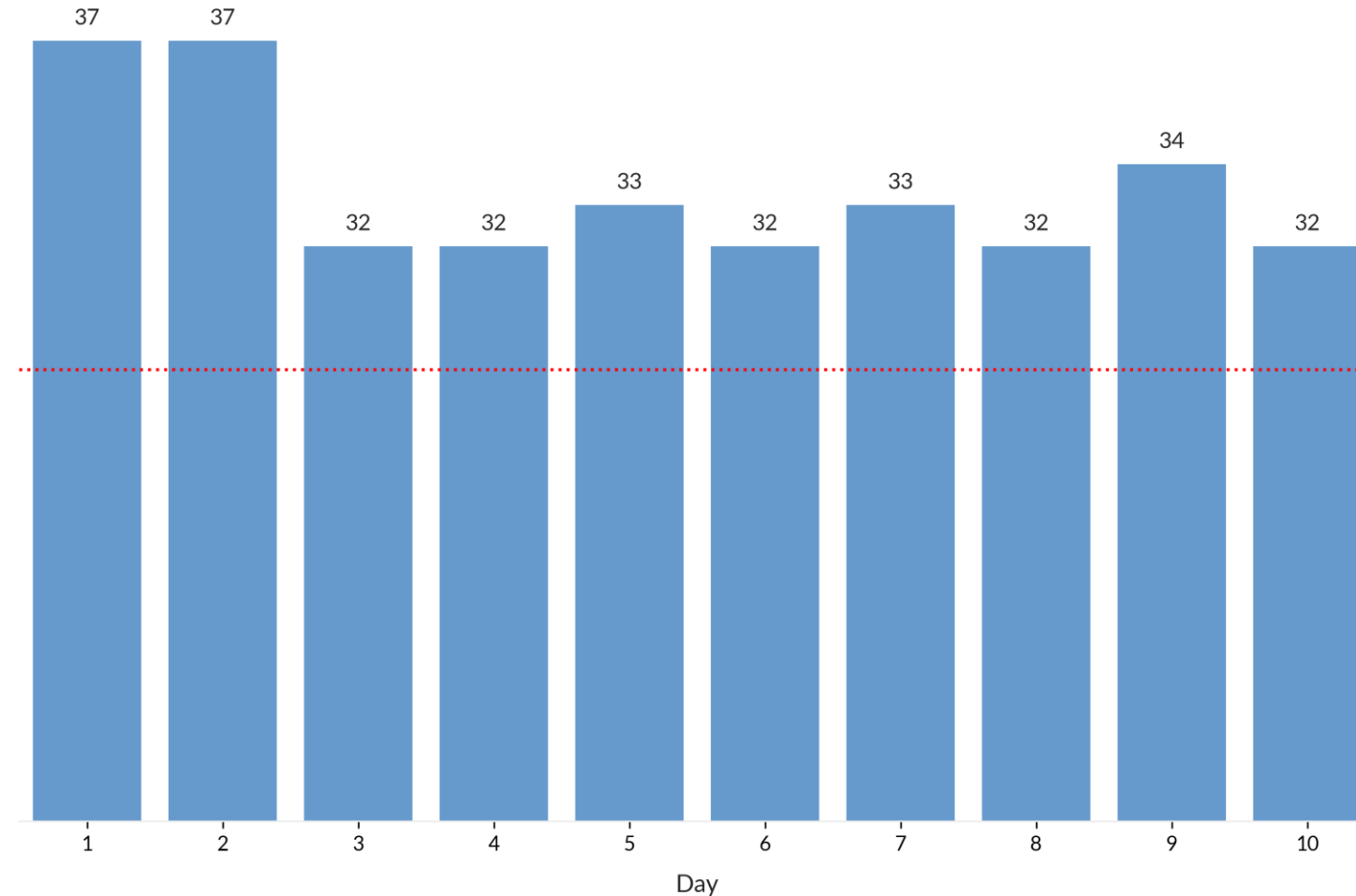
- Runs thousands of “what-if” seasons using current performance.
- Turns uncertainty into clear, decision-ready probabilities.

Per-Hole Diagnostics

- Looks at how each hole has played so far.
- Confirms whether the forecast’s assumptions match actual strengths and weak spots.

Snapshot at Day 10: Performance So Far and What It Means for the Forecast

Total Strokes per Round — through Day 10



Insights

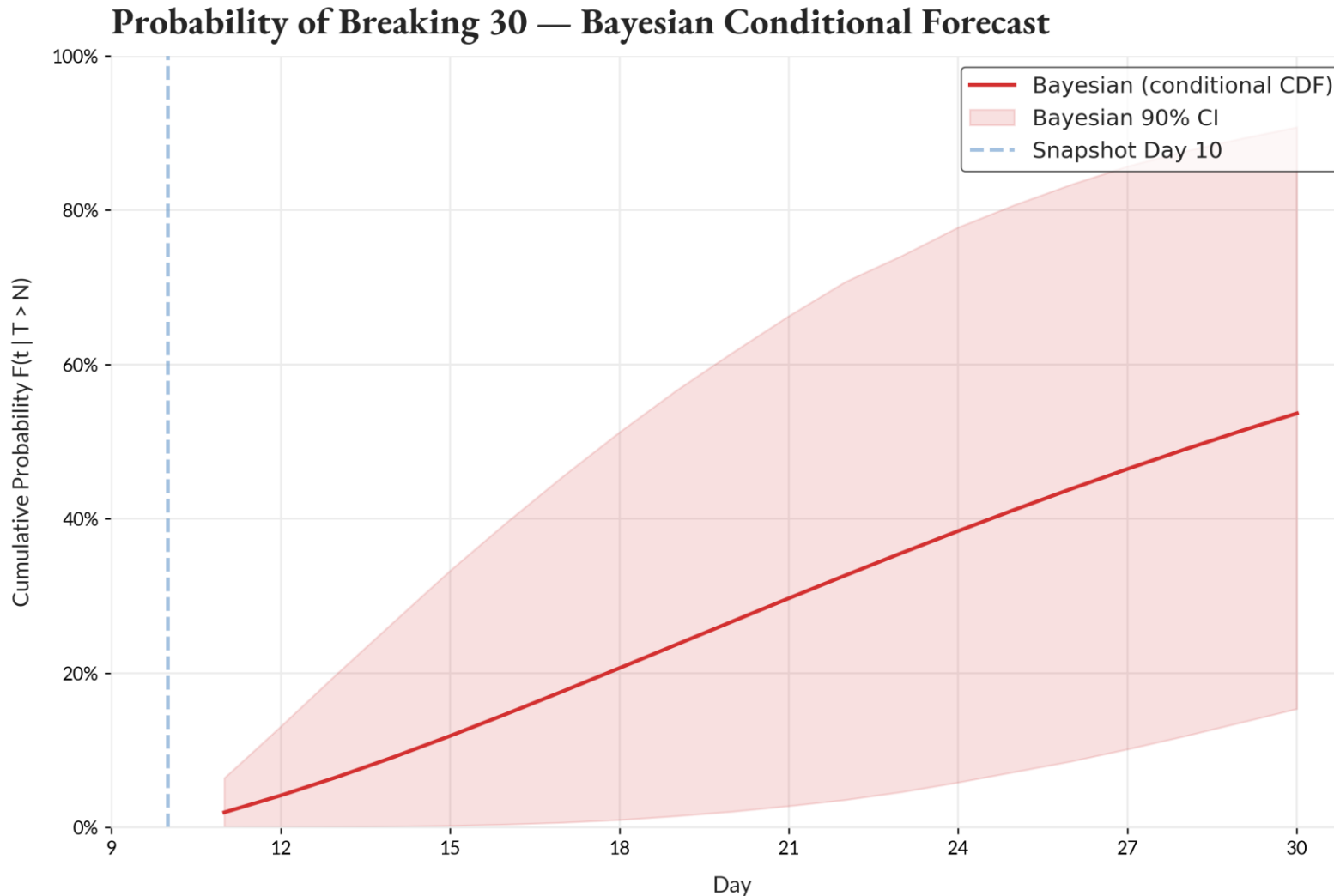
Scoring Pattern

- Opened with two 37s before settling into a steady rhythm.
- Last seven rounds clustered tightly between 32 and 34.
- No breakout yet, but consistency has clearly improved.

What It Suggests

- The margin to the milestone is still roughly three strokes.
- Sustained consistency supports a confident forecast base.
- A single hot round could flip the outlook overnight.

Bayesian Forecast: Probability of Breaking 30



Insights

Forecast Shape

- Probability climbs steadily after Day 10, reflecting consistent form but no breakout yet.
- The slope flattens near the end, signaling slower gains later in the season.

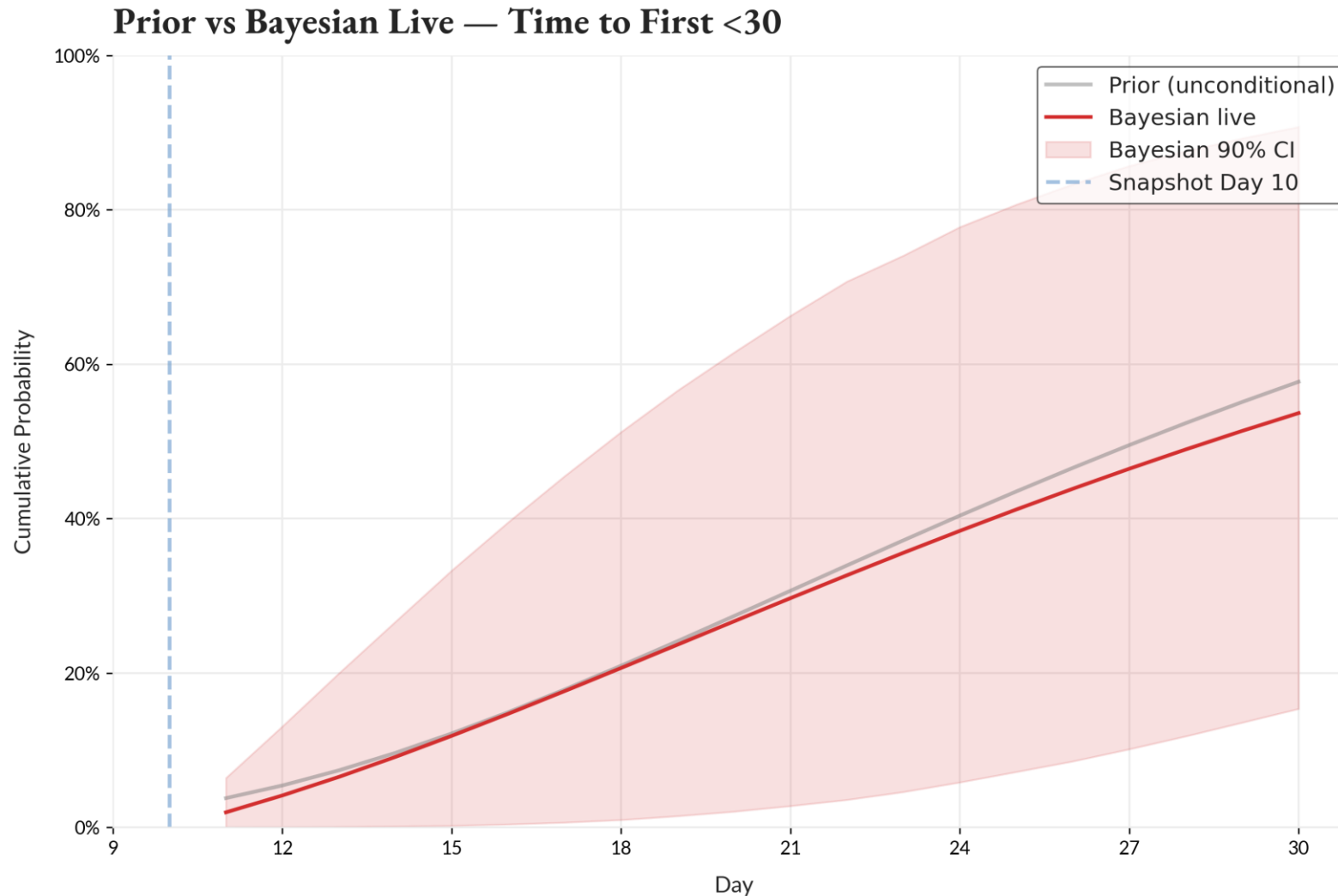
Uncertainty Range

- The 90% credible band widens with time, showing growing unpredictability.
- The model stays confident about near-term trends, less so beyond Day 20.

Key Probabilities

- ≈12% chance of breaking 30 by Day 15.
- ≈35% chance by Day 20.

Current Forecast Versus History



Insights

Relative Shape

- The live forecast (red) begins slightly below the historical prior, converging around Days 16–17.
- After that, it flattens sooner, indicating a slower late-season climb than the historical pattern.

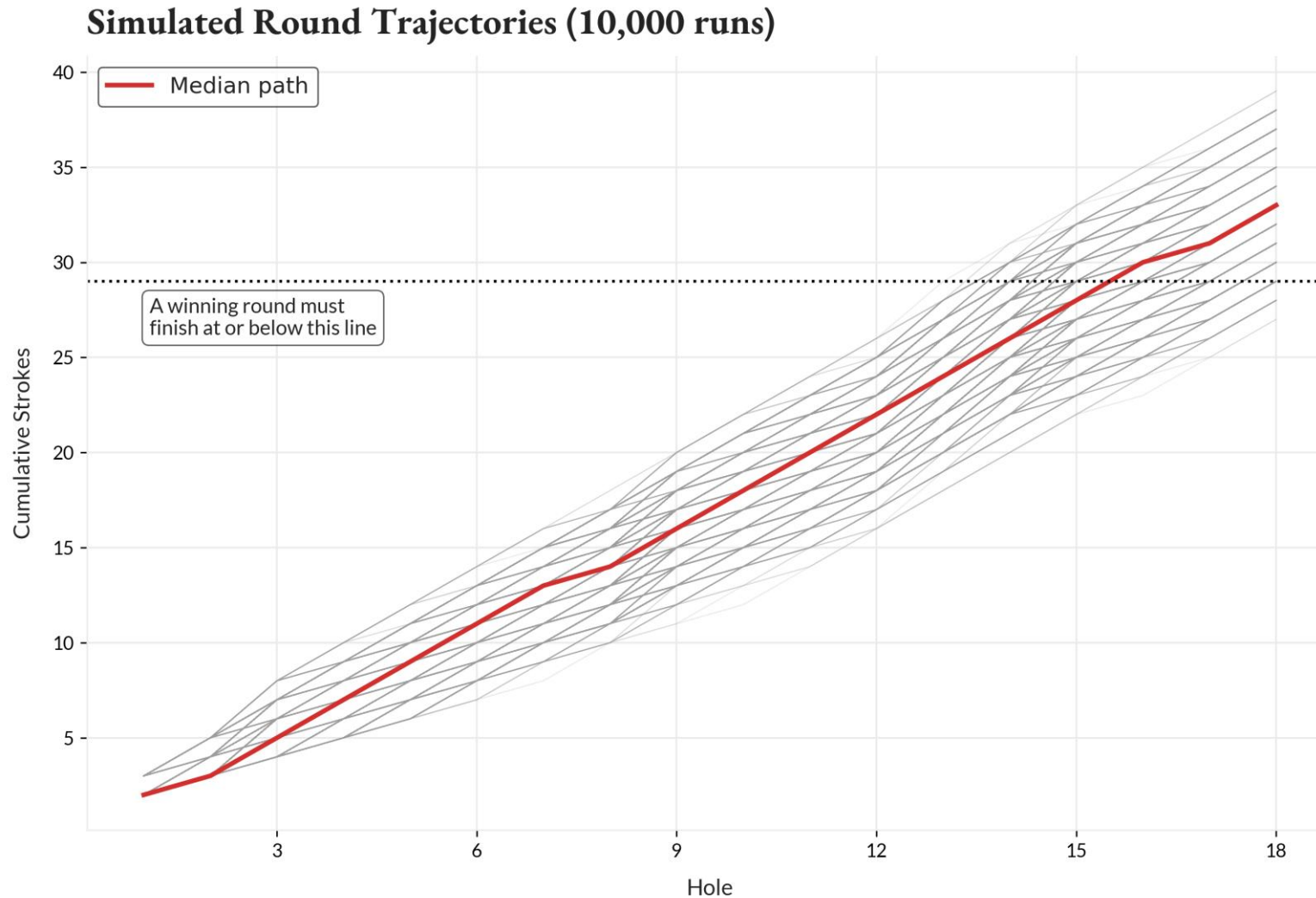
Historical Context

- Past breakthroughs typically occurred between Days 13–16, aligning with the mid-season convergence zone.
- The Day 10 trajectory reflects comparable timing but signals less acceleration beyond Day 20.

Key Probabilities

- Near-term odds remain within the expected historical envelope.

The Decision Lens: A Monte Carlo Simulation of 10,000 Possible Rounds



Insights

Simulation Results

- Each line represents one possible round drawn from current scoring probabilities.
- 0.93% of simulations finish under 30 strokes (≈ 1 in 108).

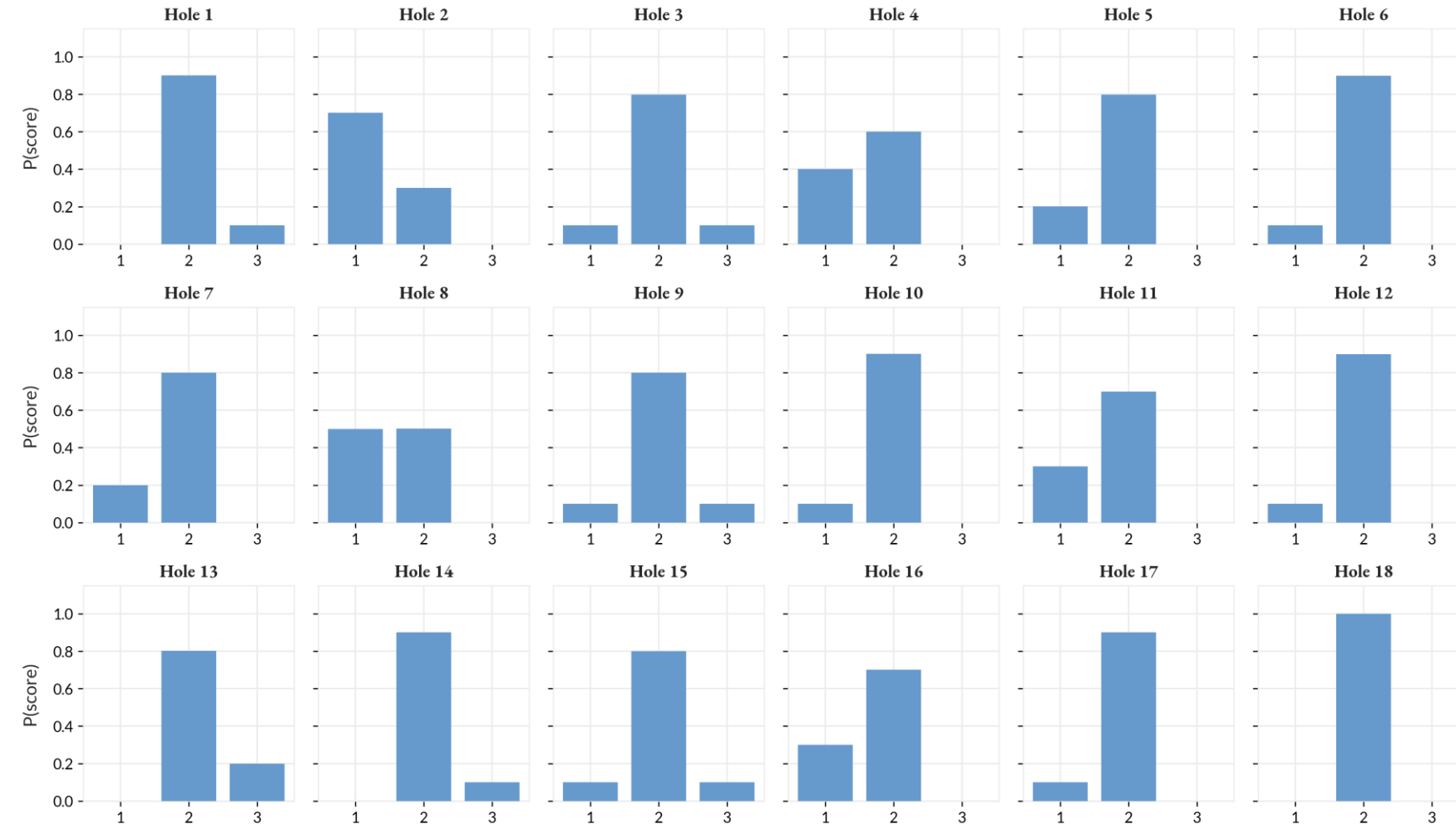
Interpretation

- Most outcomes cluster around 33–34 strokes, 3–4 above the target.
- The low tail remains thin, indicating the breakthrough remains rare but possible.

Decision Framing

- This lens models a single round, not cumulative days.

Per-Hole Diagnostics: Where the Bottleneck Lies



Insights

Performance Results

- The front nine is already quite efficient, averaging ≈ 15.6 strokes and already reaching a best of 14.
- The back nine averages about 17.4 strokes, with a best of 16, typically 1-2 over the pace needed to break 30.

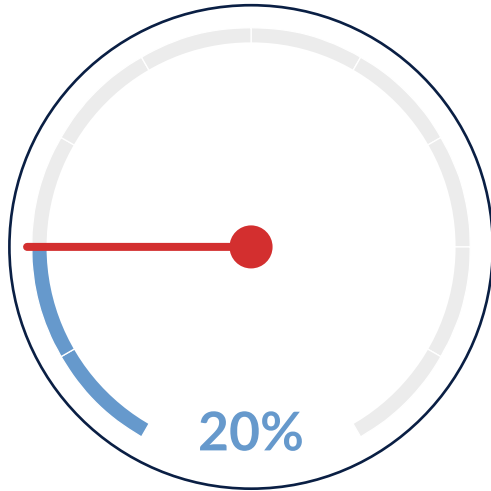
Interpretation

- Hole-level probabilities confirm the issue: aces are scarce and twos dominate on holes 13-18.

Decision Framing

- Front nine already near optimal.
- Breaking 30 likely depends on converting at least one additional ace on the back nine.

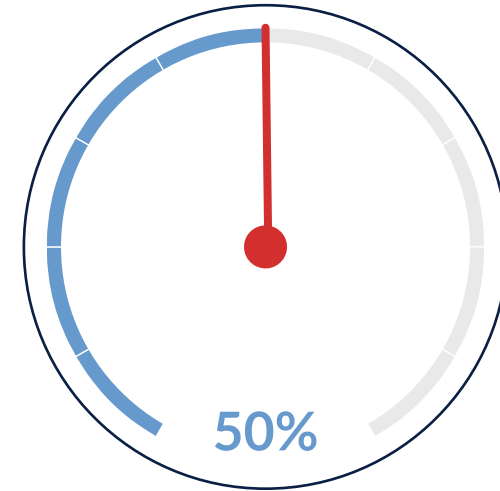
When Will They Break 30?



Day 19

Early Window Opens

The milestone becomes meaningfully possible around Day 19, when odds approach 20%. This marks the first credible point where improved execution could produce a sub-30 round.



Day 29

Median Expectation

Half of simulated outcomes reach 29 strokes or fewer by Day 29, near the 50% mark. The curve flattens afterward, showing steady progress and moderate uncertainty late in the season.

03

Business Value & Implications

*From mini-golf milestones to KPI forecasting in
business contexts.*



Acting Early Under Uncertainty Is the Advantage

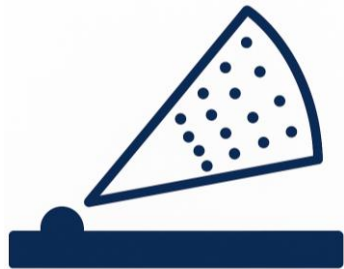


The Challenge

Rare Events Defy Easy Forecasting

Milestone events are rare and unpredictable, which makes early planning difficult. Waiting until the outcome is obvious leaves no room to adapt.

- Outcomes are binary, but timing is the real difficulty.
- Late recognition = missed opportunities.

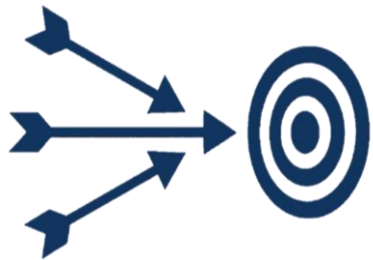


The Method

Probabilistic Tools Change the Game

Survival analysis and Bayesian updating let forecasts evolve as new data arrives. These methods turn uncertainty into probabilities decision-makers can act on.

- Forecasts update as results accumulate.
- Uncertainty is quantified, not ignored.



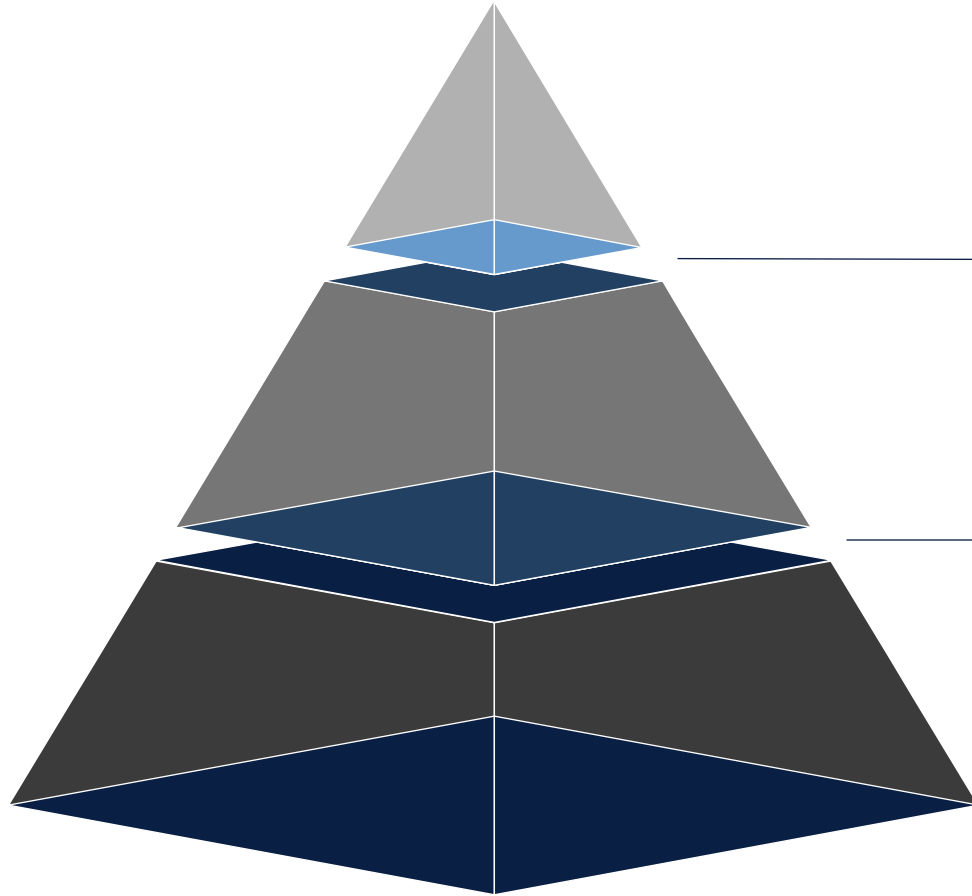
The Value

Early Signals Enable Better Decisions

Credible early forecasts move decisions from reactive to proactive. Leaders can adjust resources and prepare scenarios before thresholds are locked in.

- Resources can be reallocated sooner.
- Leaders get scenario-ready insights.

Lessons That Build Into Better Forecasting



Threshold Probabilities

Actionable estimates at critical milestones guide smarter allocation of resources and preparation for different scenarios.

Conditional Updates

Each new datapoint refines expectations, turning uncertainty into evolving probabilities. The model learns as results unfold.

Historic Baselines

Prior data provides the foundation for any credible forecast. Without history, there's no anchor for probabilities.

From Mini-Golf to Business Insight

Full Pipeline

Data Foundation

- Cleaned and validated inputs from Excel into reproducible CSVs.
- Built schema contract to lock feature definitions.
- Automated daily snapshots for live season state.

Forecasting Build

- Historical priors estimated via Kaplan–Meier and Bayesian AFT models.
- Live forecasts updated round-by-round with survival + posterior draws.
- Modular notebooks linked into a cohesive end-to-end workflow.

Credible Communication

Uncertainty Made Clear

- Posterior bands plotted to show probability ranges over time.
- Forecasts framed as probabilities, not binary predictions.
- Credible intervals visualized instead of hidden behind averages.

Actionable Forecasts

- Threshold-specific probabilities (e.g., odds of “breaking 30” by Day N).
- Scenario-ready outputs for leaders to plan against.
- Forecasts communicated as evolving signals, not static reports.

Transferable Lessons

Storytelling Skills

- Python stack: pandas, lifelines, PyMC, matplotlib.
- Bayesian hierarchical modeling with course-level random effects.
- Monte Carlo + Bayesian methods combined with survival analysis.

Storytelling Skills

- Complex uncertainty distilled into intuitive visuals.
- Narrative structure: Challenge → Method → Value → Lessons.
- Business analogies (mini-golf milestones → KPI thresholds) to bridge audiences.

Forecasting Rare Events Creates Early Action Advantage



Thank you! Full code & analysis available on GitHub.

Daniel Elmore



<https://www.linkedin.com/in/daniel-elmore/>



danieljosephelmore@gmail.com